

Partial result for Godefroy's Problem 7

The non- ℓ_2 -saturated case has countably many non-isomorphic subspaces

status: partial result, not a full solution

Source Problem

The source is Dodos–Lopez-Abad–Todorćević, *Banach spaces and Ramsey Theory: some open problems*, arXiv:1006.2668v1. Problem 7, attributed to Godefroy, asks:

Let X be a separable non-Hilbertian Banach space. Is it true that there exists a sequence (X_n) of subspaces of X which are pairwise non-isomorphic?

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Problem 7. (G. Godefroy [33]) *Let X be a separable non-Hilbertian Banach space. Is it true that there exists a sequence (X_n) of subspaces of X which are pairwise non-isomorphic?*

It is likely that Problem 7 has an affirmative answer. Indeed, several partial results have been obtained (see [27] and the references therein).

We would like to isolate an instance of Problem 7 exposing, in particular, its connection with the *Basis Problem*, a famous problem in the Geometry of Banach spaces that has received considerable attention during the 1970s and 1980s. Recall that, by the results in [55] [73], if a separable Banach space X fails to have type $2 - \varepsilon$ (or fails to have co-type $2 + \varepsilon$) for some $\varepsilon > 0$, then X contains a subspace without a Schauder basis (actually, X contains a subspace without the compact approximation property).

Problem 8. *Let X be a separable Banach space which fails to have type $2 - \varepsilon$ (or fails to have co-type $2 + \varepsilon$) for some $\varepsilon > 0$. Is it true that X contains continuum many pairwise non-isomorphic subspaces?*

It is known (see [3]) that the above problem has an affirmative answer if X is a separable weak Hilbert space not isomorphic to ℓ_2 .

7. FIXING PROPERTIES OF OPERATORS ON $C[0, 1]$

A central open problem in Banach Space Theory is the classification, up to isomorphism, of all complemented subspaces of the classical function space $C[0, 1]$.

Problem 9. (Complemented Subspace Problem for $C[0, 1]$ – see [12] [52] [63]) *Is it true that every complemented subspace of $C[0, 1]$ is isomorphic to a $C(K)$ space for*

Definitions

A separable Banach space X is ℓ_2 -saturated if every closed infinite-dimensional subspace of X contains a further subspace isomorphic to ℓ_2 .

We use the homogeneous Banach-space theorem of Komorowski–Tomczak-Jaegermann and Gowers: if a separable infinite-dimensional Banach space is isomorphic to all of its closed infinite-dimensional subspaces, then it is isomorphic to ℓ_2 .

Idea of the Proof

The obstruction is finite rather than analytic. If X did not contain a sequence of pairwise non-isomorphic subspaces, then the set of isomorphism classes of subspaces of X would be finite. Inside

any finite partially ordered family of subspace-classes, one can choose a minimal class. A representative Y of such a minimal class is isomorphic to each of its own infinite-dimensional subspaces; otherwise one would move to a strictly smaller class. The homogeneous Banach-space theorem then forces $Y \simeq \ell_2$. Repeating this inside every subspace shows that X must be ℓ_2 -saturated.

The Partial Result

Theorem 1. *Let X be a separable infinite-dimensional Banach space. If X is not ℓ_2 -saturated, then X contains a sequence of closed infinite-dimensional pairwise non-isomorphic subspaces unless X is already outside the hypothesis by being Hilbertian. In particular, every separable non-Hilbertian Banach space which is not ℓ_2 -saturated satisfies the conclusion of Godefroy's Problem 7.*

Proof. We prove the contrapositive. Suppose that X has no sequence of closed infinite-dimensional pairwise non-isomorphic subspaces. Then the set $\mathcal{C}(X)$ of isomorphism classes of closed infinite-dimensional subspaces of X is finite.

For classes $A, B \in \mathcal{C}(X)$, write $A \preceq B$ if a representative of A is isomorphic to a closed infinite-dimensional subspace of a representative of B . This is a finite preorder. Choose a minimal class A_0 and let $Y \subset X$ be a representative of A_0 . If $Z \subset Y$ is a closed infinite-dimensional subspace, then $[Z] \preceq [Y] = A_0$. By minimality, $[Z] = A_0$. Thus Y is isomorphic to all of its closed infinite-dimensional subspaces. By the homogeneous Banach-space theorem, Y is isomorphic to ℓ_2 .

Now let $W \subset X$ be any closed infinite-dimensional subspace. The same finite-class argument can be applied inside W : among the isomorphism classes represented by subspaces of W , choose a minimal one. A representative of that minimal class is isomorphic to all of its closed infinite-dimensional subspaces, hence isomorphic to ℓ_2 . Therefore every closed infinite-dimensional subspace W of X contains a further subspace isomorphic to ℓ_2 . That is, X is ℓ_2 -saturated.

Consequently, if X is not ℓ_2 -saturated, the supposition was false: X must contain a sequence of pairwise non-isomorphic subspaces. $\square$

Corollary 1. *Any counterexample to Godefroy's Problem 7 must be a separable non-Hilbertian ℓ_2 -saturated Banach space.*

Limitations and Novelty Check

This is a reduction, not a full solution. It leaves open the ℓ_2 -saturated case. It also gives only countably many pairwise non-isomorphic subspaces in the non- ℓ_2 -saturated case; it does not prove the continuum-many strengthening from Problem 8.

The run indexes were searched for arXiv:1006.2668, the survey title, Godefroy's Problem 7, and pairwise non-isomorphic subspaces. A prior attempt for a different part of the same survey, the Rosenthal/unique-spreading-model edge, was read and not duplicated. Bounded web searches for the exact ℓ_2 -saturated reduction did not reveal an obvious prior packet or later exact answer during this pass.

References

References

- [1] P. Dodos, J. Lopez-Abad, and S. Todorcevic, *Banach spaces and Ramsey Theory: some open problems*, arXiv:1006.2668v1, 2010.

- [2] W. T. Gowers, *An infinite Ramsey theorem and some Banach-space dichotomies*, Ann. of Math. (2) 156 (2002), 797–833.
- [3] R. A. Komorowski and N. Tomczak-Jaegermann, *Banach spaces without local unconditional structure*, Israel J. Math. 89 (1995), 205–226.

Human Review Recommendation

Check the finite-class/minimal-class argument and that the use of the homogeneous Banach-space theorem is exactly at the point where a minimal representative is isomorphic to all of its own infinite-dimensional subspaces. The packet should be read as a reduction of Problem 7 to the ℓ_2 -saturated case.